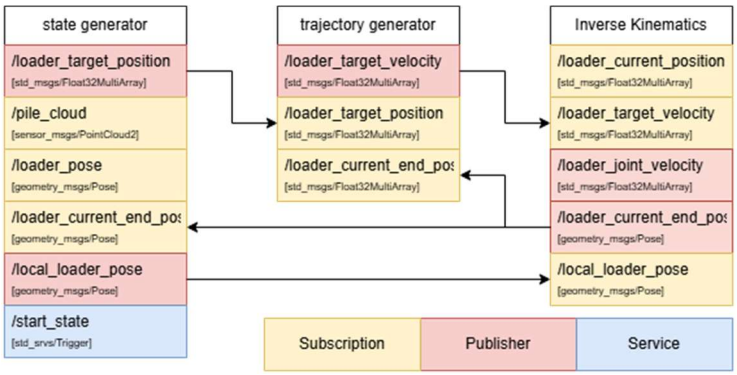
Wheel loader trajectory generation

Task description and requirements

This project is for generate trajectory for wheel loader including state and posture calculate.

- Input from LiDAR is pointcloud.
- Calculate state for move wheel loader from pile shape.
- Calculate Kinematics from task space to configuration space.
- Calculate trajectory for smooth move.

System Diagram



All system contains with 3 node

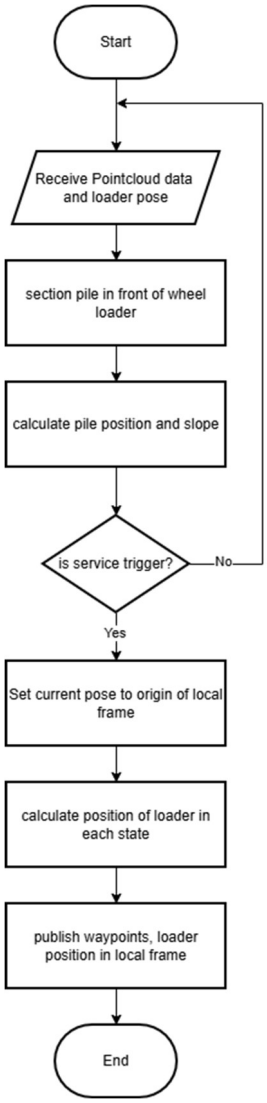
- State generator: receive pointcloud and loader position calculate pile shape and state for wheel loader. Output list of waypoints to trajectory generator.
- Trajectory generate: create smooth trajectory between each waypoint and publish target velocity.
- Inverse Kinematics: Receive task space velocity and calculate to joint space velocity of wheel loader.

Module Details

State Generator

This module get pointcloud shape and wheel loader position after get trigger it will calculate position of each state for wheel loader scoop and publish waypoint from start position to finish scoop.

Inputs: Pointcloud data, loader position, loader tip position. **Output:** waypoints(loader target position).



Slope Analysis:

- **Data Receive:** Get start pose of wheel loader and pointcloud data which have slope data from volume estimate.
- **Region Segmentation:** The algorithm filters the pointcloud to isolate ("section") only the specific pile of material located directly in front of the wheel loader.
- **Slope Determination:** using slope data of sectioned pile area and computes an average to determine the overall slope.

Process Initialization:

- **Service Activation:** The system begins by waiting for and receiving a specific service trigger to start the planning workflow.
- **Loader Local Frame:** Calculated waypoints is reference from loader current pose. After get Service trigger, it's set current pose as origin of local frame.

State Estimation:

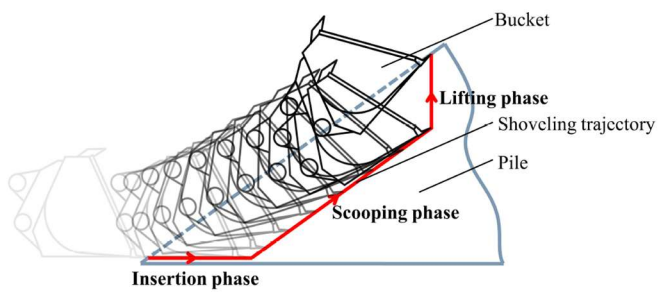


Figure 1. Shoveling trajectory.

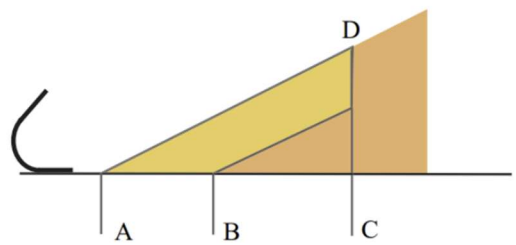


Fig.2 Bucket Motion trajectory

The algorithm separates the trajectory generation into three phases: Insertion, Scooping, and Lifting. This process calculates waypoints (x, y, theta) for critical positions A, B, C, and D.

- **Insertion Phase:** The bucket remains flat (parallel to the floor) to penetrate the pile.
  - **Point A:** Defined as the edge of the pile.
  - **Point B:** A fixed insertion depth typically set between 600 mm and 1000 mm from Point A.
- **Scooping Phase:** The bucket moves from B to C, tilting linearly to align with the pile's slope.
  - **Trajectory:** The path from B to C is calculated to be parallel to the existing pile slope.
  - **Volume Calculation:** Point C is determined such that the swept volume (represented by area ABCD) equals the bucket's total capacity * safety factor.
  - **Constraint:** The distance AC is capped at a specific limit to ensure the loader's rear wheels do not drive onto the pile.
- **Lifting Phase:** The bucket curls upward to retain the material and lifts clear of the pile.
  - **Point D:** Located directly vertical from Point C.
  - **Orientation:** The bucket angle transitions linearly from the slope angle at C to a fully curled ("all tilt") position at D.

Output:

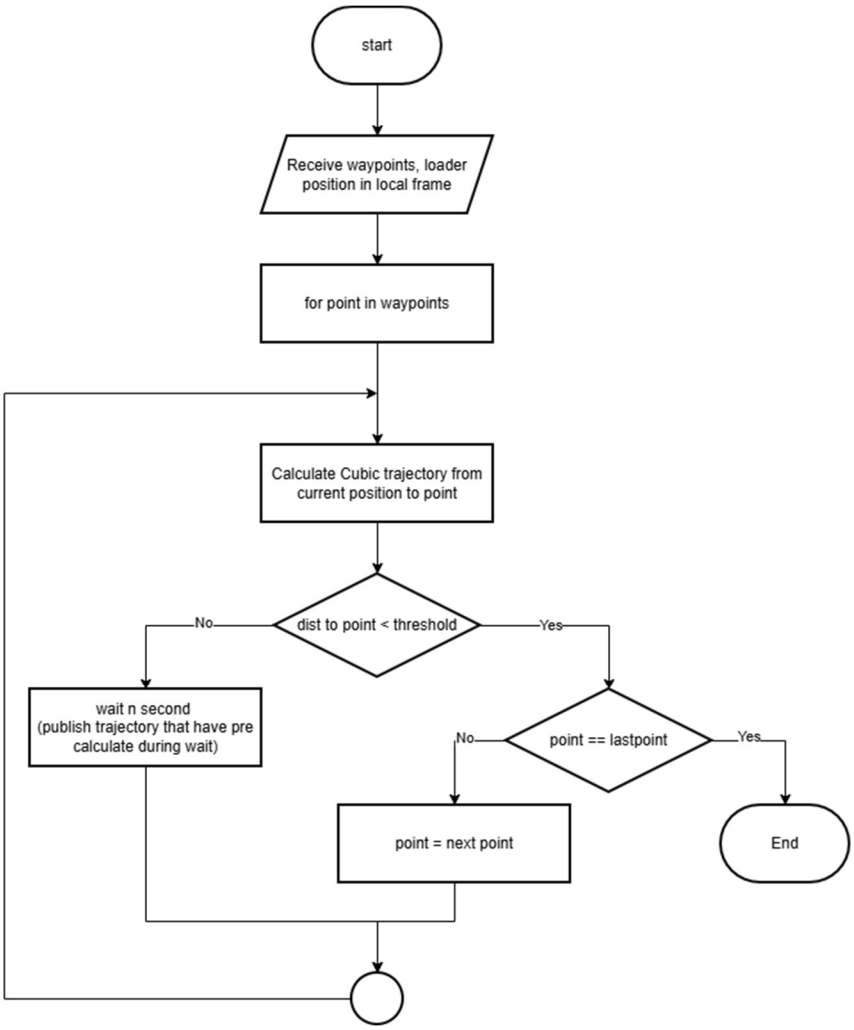
- **Waypoint Publication:** The calculated positions are formatted into navigational waypoints and published to the vehicle's control system for execution.

Trajectory Generate

This module get waypoints form state and calculate trajectory using cubic polynomial in task space and then output as velocity.

Inputs: waypoints(loader target position), current loader position. **Output:** velocity (x, y, theta).

Parameters: Vmax, Amax



Initialize and Recalculate:

- **Initialize:** The process begins by receiving a target set of waypoints and the wheel loader's current position within the local coordinate frame.

- **Dynamic Replanning:** To maintain high precision, the system executes a periodic replan every xx seconds. It calculates a new cubic trajectory from the current feedback position to the nearest upcoming waypoint.
- **Threshold check:** The controller continuously monitors the Euclidean distance to the active waypoint. Once the distance is below a defined threshold, the target automatically advances to the next point in the sequence until the final goal is reached.

Cubic Trajectory Profile:

To maintain smooth motion and protect the loader's mechanical integrity, the trajectory is solved as a 3rd-degree polynomial:

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$$

- **Coefficient Calculation (a_n):**
  - $a_0 = x_0$
  - $a_1 = v_0$
  - $a_2 = \frac{3(x_T - x_0) - T(2v_0 + v_T)}{T^2}$
  - $a_3 = \frac{2(x_0 - x_T) + T(v_0 + v_T)}{T^3}$
- **Time Calculation (T)**: The duration is determined by the most restrictive physical limit v_{max} or a_{max}.
  - **Acceleration-Limited:** Derived from the quadratic formula:

$$T_a = \frac{-(4v_0 + 2v_T) + \sqrt{(4v_0 + 2v_T)^2 + 24a_{max}|x_T - x_0|}}{2a_{max}}$$

  - **Velocity-Limited:** For a cubic profile, the velocity follows a parabolic shape. By integrating the parabola from y=0 to y=0, the average velocity is found to be around 2/3 of the peak. Therefore, we estimate the required time using the 1.5 factor to ensure the peak does not pierce v_max:

$$T_v = 1.5\ \frac{|x_T - x_0|}{v_{max}}$$

  - The system selects the final segment time by taking the maximum of both constraints: T = max(T_a, T_v).

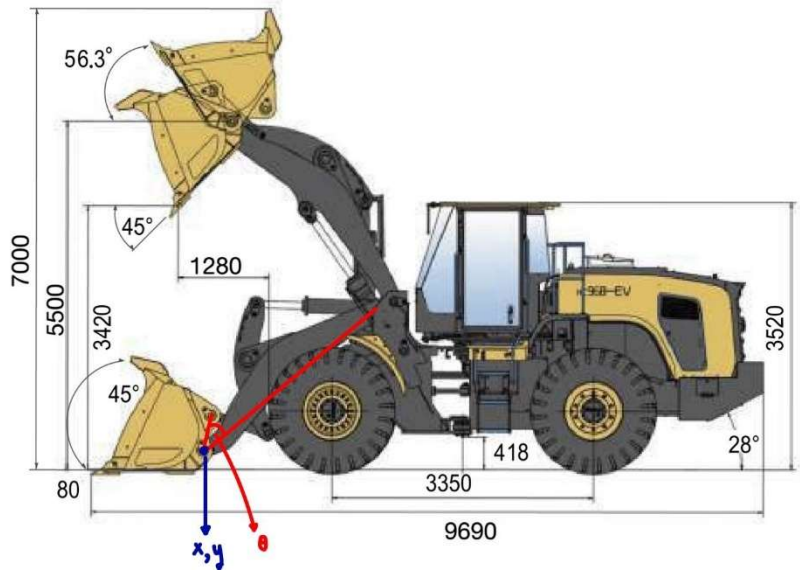
Output

The final output is a sequence of time-stamped velocity.

Kinematics

In wheel loader there are 3 control joint: wheel(w), lift cylinder(p_l), tilt cylinder(p_t)

Given x,y is position of bucket tilt joint and theta is angle between bucket and arm

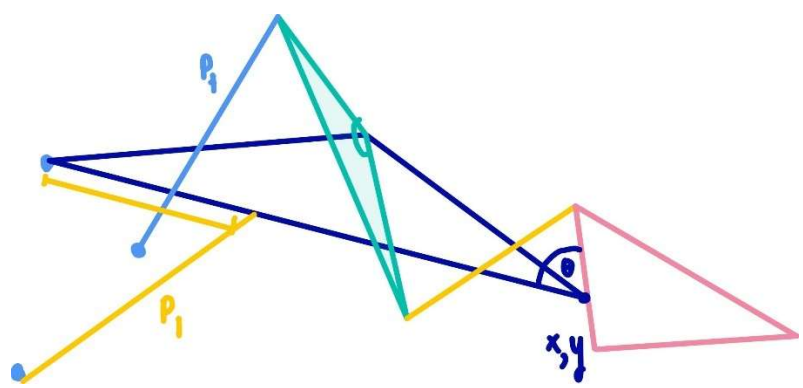


Each variable can write in function as

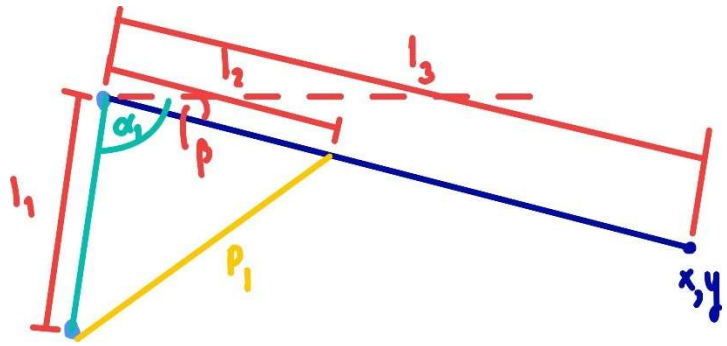
$$x = f(w, p_l)$$

$$y = f(p_l)$$

$$\theta = f(p_l, p_t)$$



For set x, y and theta position we first set pl by y position and then set w and pt by x and theta position



- Inverse Differential Kinematics of y

$$y = l_3 \sin\beta$$

$$\beta = \alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)$$

$$y = l_3 \sin\left(\alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)\right)$$

$$\dot{y} = \frac{l_3 \cos\left(\alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)\right)}{\sqrt{1 - \left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)^2}}\left(\frac{-P_l}{l_1l_2}\right)\dot{P}_l$$

$$\dot{P}_l = \left(\frac{l_3 \cos\left(\alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)\right)}{\sqrt{1 - \left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)^2}}\left(\frac{-P_l}{l_1l_2}\right)\right)^{-1}\dot{y}$$

- Inverse Differential Kinematics of x

$$x = l_3 \cos\beta + x_{wheel}$$

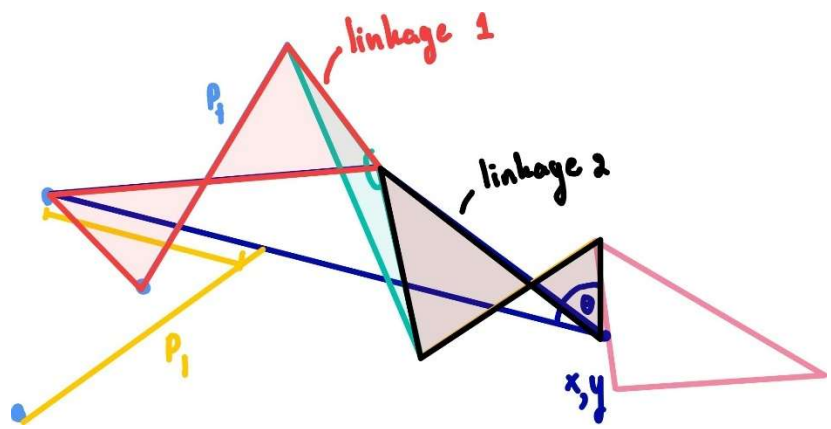
$$\beta = \alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)$$

$$x = l_3 \cos\left(\alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2l_1l_2}\right)\right) + x_{wheel}$$

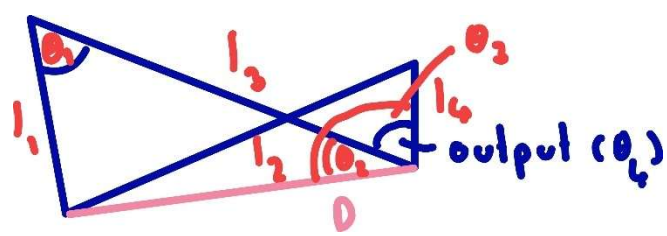
$$\dot{x} = \frac{l_3 \sin\left(\alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2 l_1 l_2}\right)\right)}{\sqrt{1 - \left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2 l_1 l_2}\right)^2}} \left(\frac{P_l}{l_1 l_2}\right) \dot{P}_l + \dot{x_{wheel}}$$

$$\dot{x_{wheel}} = \dot{x} - \frac{l_3 \sin\left(\alpha_1 + \cos^{-1}\left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2 l_1 l_2}\right)\right)}{\sqrt{1 - \left(\frac{{l_1}^2 + {l_2}^2 - {P_l}^2}{2 l_1 l_2}\right)^2}} \left(\frac{P_l}{l_1 l_2}\right) \dot{P}_l$$

- Inverse Differential Kinematics of theta



Tilt link is structure of 2 4 bar linkage so we can solve by



$$D^2 = {l_1}^2 + {l_3}^2 - 2 l_1 l_3 \cos\theta_1$$

$${l_1}^2 = D^2 + {l_3}^2 - 2 D l_3 \cos\theta_2$$

$${l_1}^2 = {l_1}^2 + {l_3}^2 - 2 l_1 l_3 \cos\theta_1 + {l_3}^2 - 2\sqrt{{l_1}^2 + {l_3}^2 - 2 l_1 l_3 \cos\theta_1}\, l_3 \cos\theta_2$$

$$\theta_2 = \cos^{-1} \frac{2{l_3}^2 - 2 l_1 l_3 \cos\theta_1}{2 l_3 \sqrt{{l_1}^2 + {l_3}^2 - 2 l_1 l_3 \cos\theta_1}}$$

$${l_2}^2 = D^2 + {l_4}^2 - 2 D l_4 \cos\theta_3$$

$$l_2{}^2 = l_1{}^2 + l_3{}^2 - 2l_1 l_3 \cos\theta_1 + l_4{}^2 - 2\sqrt{l_1{}^2 + l_3{}^2 - 2l_1 l_3 \cos\theta_1}\, l_4 \cos\theta_3$$

$$\theta_3 = \cos^{-1}\frac{l_1{}^2 + l_3{}^2 - l_2{}^2 + l_4{}^2 - 2l_1 l_3 \cos\theta_1}{2l_4\sqrt{l_1{}^2 + l_3{}^2 - 2l_1 l_3 \cos\theta_1}}$$

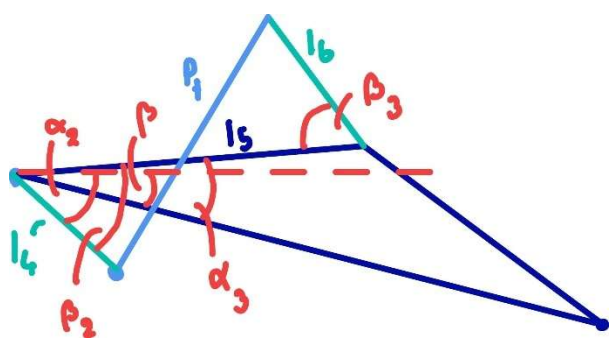
$$\theta_4 = \theta_3 - \theta_2$$

$$\theta_4 = \cos^{-1}\frac{l_1{}^2 + l_3{}^2 - l_2{}^2 + l_4{}^2 - 2l_1 l_3 \cos\theta_1}{2l_4\sqrt{l_1{}^2 + l_3{}^2 - 2l_1 l_3 \cos\theta_1}} - \cos^{-1}\frac{2l_3{}^2 - 2l_1 l_3 \cos\theta_1}{2l_3\sqrt{l_1{}^2 + l_3{}^2 - 2l_1 l_3 \cos\theta_1}}$$

Theta 4 is close form answer of 4 bar linkage

For tilt link

First linkage:

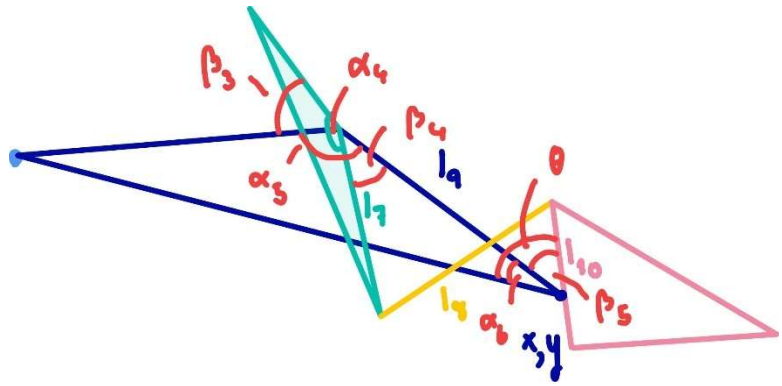


$$\beta_2 = \alpha_2 - \beta + \alpha_3, \qquad \dot{\beta}_2 = \dot{\beta}$$

$$\beta_3 = \cos^{-1}\frac{l_4{}^2 + l_5{}^2 - P_t{}^2 + l_6{}^2 - 2l_4 l_5 \cos\beta_2}{2l_6\sqrt{l_4{}^2 + l_5{}^2 - 2l_4 l_5 \cos\beta_2}} - \cos^{-1}\frac{2l_5{}^2 - 2l_4 l_5 \cos\beta_2}{2l_5\sqrt{l_4{}^2 + l_5{}^2 - 2l_4 l_5 \cos\beta_2}}$$

$$\dot{\beta}_3 = \frac{-\left(2l_6\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}\right)\left(-2P_t\dot{P}_t + 2l_4 l_5 \sin\beta_2\dot{\beta}_2\right) + \left(l_4^2 - P_t^2 + l_5^2 + l_6^2 - 2l_4 l_5 \cos\beta_2\right)\left(\frac{2l_6 l_4 l_5 \sin\beta_2\dot{\beta}_2}{\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}}\right)}{\sqrt{1 - \left(\frac{l_4^2 - P_t^2 + l_5^2 + l_6^2 - 2l_4 l_5 \cos\beta_2}{2l_6\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}}\right)^2}\left(2l_6\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}\right)^2}$$
$$+ \frac{\left(2l_5\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}\right)\left(2l_4 l_5 \sin\beta_2\dot{\beta}_2\right) - \left(2l_5^2 - 2l_4 l_5 \cos\beta_2\right)\left(\frac{2l_4 l_5^2 \sin\beta_2\dot{\beta}_2}{\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}}\right)}{\sqrt{1 - \left(\frac{2l_5^2 - 2l_4 l_5 \cos\beta_2}{2l_5\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}}\right)^2}\left(2l_5\sqrt{l_4^2 + l_5^2 - 2l_4 l_5 \cos\beta_2}\right)^2}$$

Second linkage:



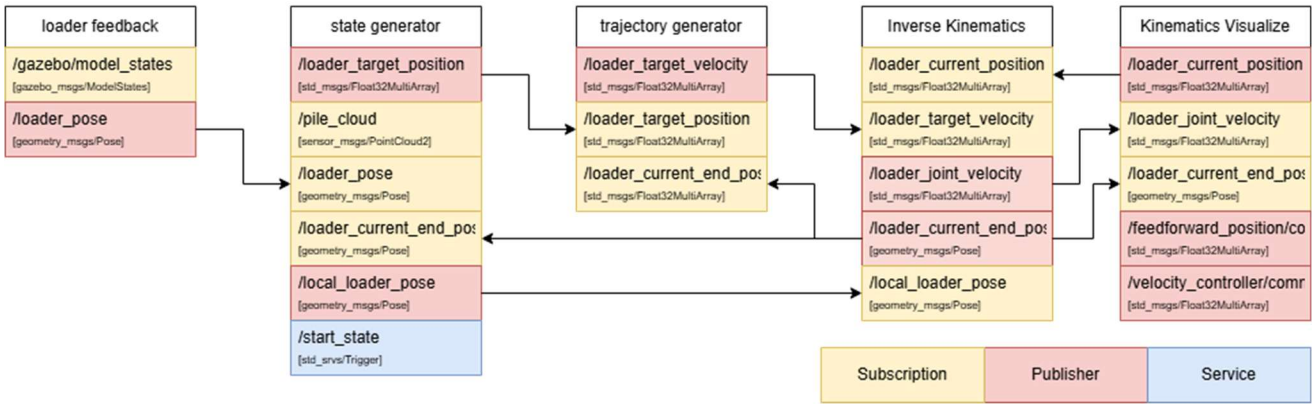
$$\beta_4 = \beta_3 + \alpha_5 - \alpha_4, \qquad \dot{\beta}_3 = \dot{\beta}_4$$

$$\beta_5 = \cos^{-1}\frac{{l_7}^2 + {l_9}^2 - {l_8}^2 + {l_{10}}^2 - 2l_7 l_9 \cos\beta_4}{2l_{10}\sqrt{{l_7}^2 + {l_9}^2 - 2l_7 l_9 \cos\beta_4}} - \cos^{-1}\frac{2{l_9}^2 - 2l_7 l_9 \cos\beta_4}{2l_9\sqrt{{l_7}^2 + {l_9}^2 - 2l_7 l_9 \cos\beta_4}}$$

$$\dot{\beta}_5 = \frac{-\left(2l_{10}\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}\right)\left(2l_7 l_9 \sin\beta_4 \dot{\beta}_4\right) + \left(l_7^2 - l_8^2 + l_9^2 + l_{10}^2 - 2l_7 l_9 \cos\beta_4\right)\left(\frac{2l_{10} l_7 l_9 \sin\beta_4 \dot{\beta}_4}{\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}}\right)}{\sqrt{1 - \left(\frac{l_7^2 - l_8^2 + l_9^2 + l_{10}^2 - 2l_7 l_9 \cos\beta_4}{2l_{10}\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}}\right)^2}\left(2l_{10}\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}\right)^2}$$

$$+ \frac{\left(2l_9\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}\right)\left(2l_7 l_9 \sin\beta_4 \dot{\beta}_4\right) - \left(2l_9^2 - 2l_7 l_9 \cos\beta_4\right)\left(\frac{2l_7 l_9^2 \sin\beta_4 \dot{\beta}_4}{\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}}\right)}{\sqrt{1 - \left(\frac{2l_9^2 - 2l_7 l_9 \cos\beta_4}{2l_9\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}}\right)^2}\left(2l_9\sqrt{l_7^2 + l_9^2 - 2l_7 l_9 \cos\beta_4}\right)^2}$$

$$\theta = \beta_5 + \alpha_6, \qquad \dot{\theta} = \dot{\beta}_5$$

Validation

For validation we add visualization to verify kinematics is correct.